

RESEARCH ARTICLE

Unconditional GAN-Based Generation of Stage-Specific Fundus Images for Augmenting Diabetic Retinopathy Screening

MICHAL KOVAC¹, PETER MACSIK¹, VERONIKA KURILOVA^{1,2}, ONDREJ STRAKA¹,
ZLATICA FELLNER², AND JARMILA PAVLOVICOVA¹

¹Faculty of Electrical Engineering and Information Technology, Slovak University of Technology in Bratislava, 812 19 Bratislava, Slovak Republic

²Faculty of Medicine, Slovak Medical University in Bratislava, 833 03 Bratislava, Slovak Republic

Corresponding author: Michal Kovac (michal.kovac1@stuba.sk)

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ABSTRACT Class imbalance in diabetic retinopathy (DR) datasets, particularly the underrepresentation of advanced stages such as proliferative DR, limits classifier performance and necessitates specialized augmentation strategies. This study presents a data augmentation approach using StyleGAN2-ADA to generate stage-specific synthetic fundus images. Unlike conditional approaches requiring labels or segmentation masks, our method trains separate unconditional models for each DR severity stage, producing anatomically realistic images while preserving pathological characteristics. We propose a comprehensive evaluation framework comprising three components: quantitative assessment using distribution-based metrics (FID, KID, IS) and nearest-neighbor-based metrics (NN-SSIM, NN-PSNR); task-based evaluation through DR classification experiments with ConvNeXt Tiny, MobileNet V3 Small, and Xception architectures; and expert validation by three ophthalmologists assessing clinical plausibility. Classification experiments demonstrated improvements in overall accuracy, with ConvNeXt Tiny achieving 73.48% accuracy (+1.82% over baseline) when combining synthetic and classical augmentation. Inter-rater reliability analysis between the two most discriminating evaluators ($\kappa = 0.723$) confirmed the clinical realism of generated images. Our unconditional StyleGAN2-ADA approach provides a scalable, mask-free augmentation solution addressing class imbalance in fundus imaging datasets, potentially enhancing DR screening systems in resource-constrained clinical settings.

INDEX TERMS Diabetic retinopathy, retinal image synthesis, synthetic medical images, generative adversarial network, deep learning, DR grading.

I. INTRODUCTION

Diabetic retinopathy (DR) is a microvascular complication of diabetes and a leading cause of preventable blindness worldwide. Early detection and timely intervention are crucial for preserving vision; however, manual screening by ophthalmologists is resource-intensive and prone to human variability. Deep learning offers promising automation for

DR screening from color fundus images, potentially enabling faster triage and reducing clinician workload [1], [2], [3], [4], [5]. However, the effectiveness of these automated systems is critically dependent on the availability of large, balanced training datasets.

Diagnosis of DR requires recognizing subtle pathological changes in multiple stages of the disease, from mild non-proliferative DR (NPDR) to vision-threatening proliferative DR (PDR). Neural networks have proven highly effective for image classification and segmentation in medical

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imaging [6], [7], [8], [9], [10], [11]. However, training high-performance models requires not only large sample sizes but also balanced class distributions. Medical datasets often exhibit severe imbalance, making it challenging to detect early stages. In contrast, advanced stages are rare due to effective disease management or limited patient access, and they can appear heterogeneous. This imbalance causes models to achieve high accuracy by simply predicting majority classes while failing to recognize critical minority classes like PDR.

Creating comprehensive medical datasets is both time-consuming and expensive, requiring expert clinicians to examine patients, technicians to acquire images, and specialists to provide accurate labels, often requiring consensus from multiple experts. Additionally, patient privacy regulations complicate data sharing and aggregation across institutions, further limiting the availability of datasets.

This paper addresses class imbalance in DR datasets by generating synthetic images using Generative Adversarial Networks (GANs). While classical augmentation techniques (rotations, flips, brightness adjustments) [12] provide some benefit, they only create variations of existing images without introducing genuinely novel anatomical or pathological patterns. Recent advances in generative models, particularly GANs [13], which have demonstrated success in medical image synthesis [14], [15], [16], offer the potential to generate entirely new, realistic images that expand the effective training distribution.

Our approach explicitly targets the challenge of GAN-based generation for imbalanced DR stages. We train separate StyleGAN2-ADA models for each class, ensuring accurate representation of stage-specific pathological features while avoiding potential label confusion during training. This unconditional approach contrasts with Conditional GANs (CGANs) [17], which have been used to produce synthetic medical data [18], [19] but require explicit conditioning signals and often lesion segmentation masks. Although masks can guide the generation of specific pathologies, they limit the diversity of output and require extensive manual annotation. StyleGAN2-ADA [20], [21] offers high training stability and sophisticated style control through its mapping network, enabling the generation of diverse anatomically accurate fundus images without explicit conditioning or mask requirements.

The main contributions of this work include a practical, unconditional training approach using StyleGAN2-ADA to generate realistic, stage-specific fundus images from imbalanced datasets, without requiring lesion masks or conditional labels. The goal was to test our synthetic augmentation with unconditionally generated images alone and in combination with classical augmentation in DR classification, analyzing performance, particularly for minority classes, compared with performance with or without classical augmentation techniques. We aimed to perform our experiments on the IDRiD dataset, which was also used for image generation,

and on the FGADR dataset, which was not used for image generation. Additionally, we propose a comprehensive, multi-step evaluation pipeline for synthetic medical images, which we apply for the first time to our StyleGAN2-ADA image-generation approach.

II. RELATED WORK

Diabetic retinopathy is typically assessed from color fundus photographs, and most computer vision pipelines for DR data focus on three main tasks: image-level grading, lesion-level segmentation, and structure analysis. CNN-based classifiers trained on large-scale datasets [8], such as EyePACS and FGADR, have achieved strong performance in predicting DR severity grades. However, their accuracy degrades for underrepresented stages, including mild NPDR or PDR, where only a small number of labeled images are available [22]. Complementary segmentation datasets such as IDRiD [23], REFUGE [24], RIMONE_DL [25], Drishti-GS [26], DRIVE [27], STARE [28], CHASE DB1 [29] and HRF [30] provide pixel-level annotations for lesions, vessels, optic disc/cup and macular structures, enabling joint classification and segmentation pipelines and more interpretable clinical models [31], [32], [33]. In all these settings, the performance of deep models is strongly constrained by dataset size, label quality, and class imbalance, especially the scarcity of advanced DR cases, which motivates methods that can generate realistic, stage-specific fundus images to augment training data for both classification and segmentation.

A. GAN AND StyleGAN

Generative adversarial networks (GANs) [13] are a widely used family of deep generative models, in which a generator learns to synthesize images from noise. In contrast, a discriminator learns to distinguish between real and generated samples. Through the resulting adversarial game, the generator gradually approximates the distribution of real images. Subsequent work has focused on improving stability and output resolution. Progressive Growing of GANs (PGGAN) [34] introduced a coarse-to-fine training schedule, in which both the generator and discriminator are gradually expanded to higher resolutions, thereby significantly improving high-resolution synthesis.

StyleGAN further redesigned the generator architecture to achieve better control over image attributes [20]. Instead of feeding the latent vector directly into the generator, a mapping network produces an intermediate style representation that modulates each convolutional block via adaptive instance normalization (AdaIN). Additional per-layer noise enables fine-grained stochastic variation. This style-based architecture disentangles coarse and fine factors of variation and has become a standard choice for high-fidelity image synthesis. In medical imaging, StyleGAN and its successors (e.g., StyleGAN2 with adaptive discriminator augmentation, ADA) are particularly attractive because they can produce

high-resolution, diverse samples even from relatively small datasets, which is critical in clinical applications.

Compared to GANs, diffusion-based models have demonstrated improved robustness and anatomical realism in medical image generation, albeit with significantly higher inference and training costs, positioning GAN-based methods as a computationally efficient alternative in resource-constrained scenarios [35], [36], [37].

B. SYNTHESIZING RETINAL IMAGES

Because labeled retinal images are costly to acquire and often heavily imbalanced across disease stages, there has been growing interest in synthesizing fundus images for data augmentation and privacy-preserving sharing. We focus here on works that explicitly generate fundus or closely related retinal images using GANs or diffusion models.

Kim et al. [38] developed a synthetic fundus image generation pipeline based on a denoising diffusion probabilistic model (DDPM). They trained the model on 1,000 low-resolution (128×128) fundus images and compared it to a PGGAN baseline. Two ophthalmologists assessed the generated samples and found that the DDPM avoided mode collapse and produced diverse images. Still, the samples were less sharp than those produced by PGGAN, especially around the optic nerve and vascular tree. Computational resources were limited, higher-resolution synthesis (256×256) was not available, and the images were not yet suitable for direct clinical use, although they are promising for downstream segmentation-oriented tasks.

Shenkut and Bhagavatula [39] proposed FundusGAN, a two-stage GAN-based model. First, a patch-level U-Net (PUNet-33) segments the retinal vasculature; then, an unsupervised image-to-image translation network generates full fundus images from the vessel trees. They evaluated segmentation performance and showed that classifiers trained on a mixture of real and synthetic images benefited from FundusGAN data, particularly in data-scarce regimes. The approach was validated on DRIVE [27], CHASE DB1 [29], STARE [28], and HRF [30]. While the method produces privacy-preserving synthetic data, capturing the finest vessels and subtle structures consistently remains challenging.

Zhou et al. [19] introduced DR-GAN, a conditional GAN tailored to generate DR fundus images with fine-grained, lesion-specific control (e.g., microaneurysms, exudates). A conditional generator receives lesion-type and severity signals and produces stage-specific synthetic images, optimized with a conditional loss to enforce both realism and lesion correctness. They reported strong FID scores (4.53) and expert validation, and showed that augmenting EyePACS [40] and FGADR [22] with DR-GAN samples improved DR classification (accuracy +1.08%, Kappa +1.15%). The primary limitation is the model's complexity and data requirements, which necessitate large annotated datasets and substantial computational resources.

Go et al. [41] proposed a diffusion-based model to generate structurally realistic fundus images with anatomically

accurate arterial and venous trees. The method first generates arterial/venous masks and then trains a diffusion model on them to synthesize high-resolution retinal images. Qualitative Turing tests with five ophthalmologists and quantitative vessel-segmentation experiments on STARE and DRIVE demonstrated that the synthetic images were often indistinguishable from real ones in terms of vascular structure. Augmenting training data with 3,000 synthetic images improved vessel segmentation AUC by 1.1% and ACC by 1.3%. Compared to StyleGAN2 (FID 122.8), the diffusion model achieved a better FID of 86.78, but at the cost of higher computational complexity.

Liu et al. [42] presented a fine-grained multimodal diffusion framework focused on glaucoma, combining textual prompts, fundus images, and vascular segmentation maps to synthesize glaucoma-specific fundus images. By fine-tuning Stable Diffusion on these multimodal inputs, they produced high-resolution samples that improved glaucoma classification when used for augmentation. They validated their approach on REFUGE [24], RIMONE_DL [25], and Drishti-GS [26]. The main drawback is again the high computational cost of training and sampling from large diffusion models.

Kim et al. [43] focused on enhancing the quality of existing fundus images rather than synthesizing them from scratch. Using a diffusion bridge model, they improved images acquired under suboptimal conditions by denoising and contrast enhancement. The method achieved a PSNR of 28.07, an FID of 6.406, and an IoU of 0.926, and notably improved downstream segmentation and classification performance on private medical datasets. However, it still requires substantial training data and is primarily targeted at image restoration rather than de novo synthesis.

Alimanov and Islam [44] proposed a multi-stage DDPM pipeline for generating high-quality synthetic retinal images to improve vessel segmentation. First, a DDPM is trained to create vessel trees from noise; a second DDPM then generates fundus images conditioned on vessel label maps. Subsequent super-resolution and segmentation stages refine and validate the images. Compared with a DCGAN baseline (FID 162.50), their diffusion-based method achieved a substantially lower FID of 48.45. Experiments on DRIVE and synthetic datasets showed that the generated images improved segmentation accuracy, sensitivity, and F1-score. The main limitation is the long training time and the complexity of the multi-stage pipeline.

Collectively, these works show that both GAN- and diffusion-based models can generate realistic, clinically meaningful retinal images that support classification, segmentation, and restoration tasks. Nevertheless, most existing methods rely on conditional architectures, explicit vessel masks, or multimodal inputs and typically assume access to large, well-annotated datasets. This restricts their applicability in settings where only small, imbalanced datasets are available, or where mask annotation is infeasible.

Motivated by these limitations, we explore a simpler alternative: unconditional image generation using the StyleGAN framework. Instead of conditioning a single generator on labels or masks, we train separate StyleGAN2-ADA models for each DR stage and ask whether high-quality, stage-specific fundus images can be generated directly from noise, even under strong class imbalance.

C. DATA AUGMENTATION IN MEDICAL IMAGING: CLASSIC VERSUS GENERATIVE

Classical image-space augmentation (affine transforms, random crops, flips, photometric jitter, noise) preserves labels while introducing local appearance variability. These operations are effective regularizers but are fundamentally limited: they do not alter the global structure or lesion co-occurrence statistics, and thus provide only modest coverage of rare phenotypes in small, imbalanced datasets. In contrast, generative approaches learn class-conditional or per-class data distributions and can synthesize novel, anatomically plausible configurations that enrich the support of minority classes. GANs remain a popular choice for high-fidelity synthesis and domain augmentation; diffusion and transformer-based generators provide powerful alternatives but typically incur higher training and sampling costs.

Recent surveys in medical imaging report that deep generative augmentation (GANs, VAEs, diffusion, transformer-augmented generators) can yield richer, task-relevant variability than classical transforms, particularly under data scarcity, and emphasize the need for modality-specific augmentation policies and careful evaluation [45], [46]. Benchmarking studies on fully synthetic datasets for radiography and brain CT show that synthetic-only training can approach real-data performance in constrained label spaces but degrades with many classes or very few samples per class, and that realism improves with resolution [47]. From a model-class perspective, diffusion methods mitigate adversarial instabilities and offer broad mode coverage [41], [42], [44], [48], but their sampling is computationally intensive; modern GANs such as StyleGAN2 with ADA provide a favorable stability, compute trade-off in limited-data regimes, and transformer-augmented GANs can further improve FID with moderate memory demands [45], [46], [47], [49], [50].

These observations inform our design choices: we adopt unconditional, per-class StyleGAN2-ADA training with ADA regularization, and we evaluate our synthetic data using both standard synthesis metrics (FID) and its impact on DR classification performance. In this way, our work aligns with current best practices for safe and effective generative augmentation in medical imaging, while targeting the specific challenge of stage-specific data imbalance in DR.

III. PROPOSED METHOD

We present a methodology for generating high-quality stage-specific fundus images using StyleGAN2-ADA [21]. Rather than employing a single conditional generator, we train

separate models for each DR class to ensure precise capture of class-specific pathological characteristics while maintaining image diversity.

A. DATASETS

The IDRiD database [23] from the eye clinic in Nanded, India, contains multiple sub-datasets. We used a subset for DR disease stage classification, comprising 516 high-quality images. In all subjects, mydriasis of the patient's eye (i.e., pupil dilation) was performed before imaging. All images were recorded with the same fundus camera: the Kowa VX-10 α , with a resolution of 4288×2848 pixels and a field of view (FOV) of 50° . Hence, the images do not vary significantly in color or retinal size. However, the dataset has limitations, including a limited number of images and class imbalance, which may bias the classifier toward majority classes. Despite these limitations, the dataset is suitable for generating synthetic images of the eye's retina, given the consistent quality and acquisition conditions. The dataset is divided into training and test sets for classification. The training set consists of 413 images (80% of the total), while the test set comprises 103 images (20% of the total). The images were graded on a scale from 0 (no DR) to 4 (proliferative DR) based on the international severity scale [51]. This dataset was used for StyleGAN2-ADA training. After image generation, the training split was also balanced with synthetic images, and classification performance was evaluated. The Fine-Grained Annotated Diabetic Retinopathy (FGADR) dataset [22] provides a larger and more diverse collection of fundus images for DR classification. The dataset comprises 1,842 training images and lacks a publicly available test set; annotations follow the same international DR severity scale (0-4) as IDRiD. Images in FGADR were collected from multiple sources and exhibited greater variability in image quality, resolution, and acquisition conditions compared to IDRiD. Importantly, FGADR was not used for StyleGAN2-ADA training, making it ideal for evaluating the generalizability of synthetic images generated from the IDRiD dataset. The different class distribution in FGADR, with Severe DR being most prevalent rather than No DR or Moderate DR, as in IDRiD, provides a robust test of whether our synthetic augmentation approach transfers across datasets with different imbalance patterns. This cross-dataset validation is crucial for evaluating the practical applicability of our method in real-world clinical settings, where data characteristics vary across institutions. This dataset was balanced using synthetic images generated exclusively from the IDRiD dataset, followed by an evaluation of classification performance to assess generalization.

1) DATA PREPROCESSING

All images were preprocessed before training the neural network to ensure a consistent input size and improve model performance. Since neural networks require fixed input

dimensions, all images were resized to 512×512 from their original resolution. However, direct resizing introduced distortions in some cases due to off-center retinas. We removed the black background and centered the retina using computer vision techniques. Images were converted to grayscale, and Canny edge detection [52] was applied to extract retinal contours. A contour-detection algorithm identified the retina's bounding box, which was then used to crop and center the image. This process reduced background noise while preserving relevant features for training.

B. IMAGE GENERATION

We leverage StyleGAN2-ADA [21] for synthetic fundus image generation due to its demonstrated ability to produce high-fidelity images with substantial intra-class diversity, even from limited datasets. Unlike conditional GANs that require explicit masks or labels, our unconditional approach trains independent StyleGAN2-ADA models for each DR severity stage, offering three key advantages:

TABLE 1. Relevant hyperparameters for StyleGAN training.

Hyperparameter	Value
Batch size	8
Learning rate (G and D)	0.0025
Adam optimizer betas	[0, 0.99]
Regularization (R1 gamma)	6.5536
Total training iterations (kimg)	800
ADA target	0.6
Data augmentation	Enabled (xflip, rotate90, scale, etc.)
Resolution	512

First, each model specializes in the pathological markers of a single disease stage, ensuring accurate representation of stage-specific features without the risk of label confusion. **Second**, training independent models mitigates mode collapse by operating within distinct, well-defined distributions rather than attempting to cover all classes with a single generator. **Third**, removing explicit conditioning eliminates annotation-related biases and improves generalizability, as the model learns the full distribution of each class rather than being constrained by specific conditioning signals.

The training pipeline (Fig. 1) begins with preprocessing, which includes resolution standardization to 512×512 pixels and intensity normalization. Images are stratified by DR severity, and each subset trains a dedicated StyleGAN2-ADA model. We employ adaptive discriminator augmentation (ADA) [21] to dynamically adjust the augmentation strength, thereby preventing overfitting, particularly when training samples are limited.

Given the high computational demands of StyleGAN training, the hyperparameters were manually tuned with limited iterations to balance performance and resource utilization. The relevant parameters were carefully selected based on preliminary experiments and practical constraints.

Table 1 summarizes the manually chosen StyleGAN hyperparameters.

This approach minimizes computational overhead while achieving satisfactory image synthesis quality.

C. QUANTITATIVE EVALUATION OF GENERATED IMAGES

Quantitative comparisons of synthetic images were performed using distribution-based metrics (Fréchet Inception Distance, Inception Score, and Kernel Inception Distance) and nearest-neighbor-based pairwise metrics (nearest-neighbor Structural Similarity Index and nearest-neighbor Peak Signal-to-Noise Ratio), which assess structural and pixel-level similarity to real images.

The Fréchet Inception Distance (FID) was computed [53] between real and synthetic distributions:

$$\text{FID}(r, g) = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \quad (1)$$

where μ_r , Σ_r and μ_g , Σ_g represent the mean and covariance of feature representations for real and generated samples, respectively, FID provides robust discriminability and computational efficiency while detecting mode collapse [54]. Lower FID values indicate greater similarity between distributions.

Kernel Inception Distance (KID) measures distributional similarity using Maximum Mean Discrepancy (MMD) with a polynomial kernel [55]. Unlike FID, KID is more robust for smaller sample sizes, making it particularly suitable for medical imaging datasets with limited data. Let $X = \{x_i\}_{i=1}^n$ denote the set of feature vectors extracted from real images and $Y = \{y_j\}_{j=1}^m$ the set of feature vectors extracted from synthetic images.

$$\text{KID} = \text{MMD}^2(X, Y) \quad (2)$$

$$\text{MMD}^2 = \mathbb{E}_{x, x'}[k(x, x')] + \mathbb{E}_{y, y'}[k(y, y')] - 2\mathbb{E}_{x, y}[k(x, y)] \quad (3)$$

Similar to FID, lower KID values indicate greater similarity between distributions, with KID being preferred when sample sizes are limited.

Inception Score (IS) [56] evaluates both image quality and diversity by measuring the KL divergence between the conditional and marginal class distributions predicted by an Inception network. Let x denote a synthetic image and $p(y|x)$ the network's output distribution.

$$\text{IS} = \exp(\mathbb{E}_x[\text{KL}(p(y|x) || p(y))]) \quad (4)$$

$$p(y) = \mathbb{E}_x[p(y|x)] \quad (5)$$

Higher IS values indicate higher image quality and greater variability across images, suggesting that the generator produces diverse, realistic outputs.

The Nearest-Neighbor Structural Similarity Index (NN-SSIM) measures the structural similarity between each synthetic image and its closest real-world neighbor, focusing on the preservation of structures, textures, and local patterns.

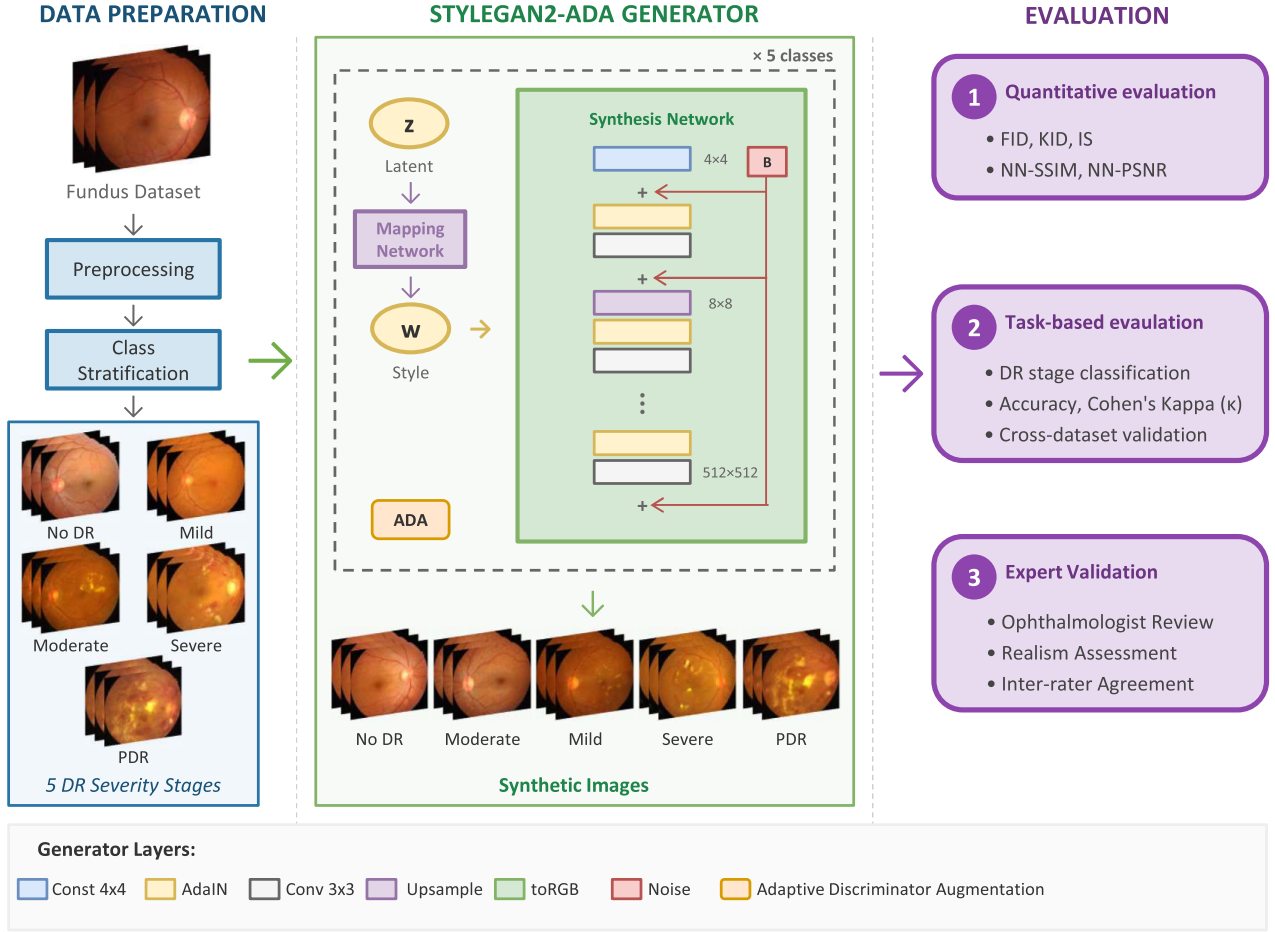


FIGURE 1. Schematic overview of the proposed methodology for GAN-based diabetic retinopathy image augmentation. Fundus images are preprocessed and stratified by DR severity (No DR, Mild, Moderate, Severe, PDR). Five independent StyleGAN2-ADA models, one per class, are trained to capture class-specific pathological features. Synthetic image quality is assessed via a three-pronged evaluation: (1) quantitative evaluation using image fidelity metrics including FID, KID, IS, NN-SSIM, and NN-PSNR; (2) task-based evaluation through DR stage classification, accuracy, Cohen's Kappa, and cross-dataset validation; and (3) expert validation by ophthalmologists measuring realism and inter-rater agreement.

For each synthetic image s , we extract a feature vector $\phi(s)$ from the pool layer of a pretrained Inception-V3 network. The nearest real image r^* is then identified using Euclidean distance in the feature space:

$$r^* = \arg \min_{r \in R} \|\phi(s) - \phi(r)\|_2 \quad (6)$$

$$\text{NN-SSIM} = \frac{1}{N} \sum_{i=1}^N \text{SSIM}(s_i, r_i^*) \quad (7)$$

Higher NN-SSIM values indicate better structural similarity between synthetic and real images. Nearest-Neighbor Peak Signal-to-Noise Ratio (NN-PSNR) similarly measures pixel-level similarity between synthetic images and their nearest real neighbors, where r^* is identified using the exact Inception-V3 feature-space matching as for NN-SSIM. We first compute the PSNR between a synthetic image s and its nearest real neighbor r^* , then average across all synthetic

images:

$$\text{PSNR}(s, r^*) = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}(s, r^*)} \right) \quad (8)$$

$$\text{NN-PSNR} = \frac{1}{N} \sum_{i=1}^N \text{PSNR}(s_i, r_i^*) \quad (9)$$

Higher NN-PSNR values indicate lower differences in pixel intensities between synthetic and real images.

D. TASK-BASED EVALUATION OF GENERATED IMAGES

We investigate whether GAN-generated synthetic data improves DR classification when used alone, in combination with classical augmentation, or as a baseline with no augmentation. Given the significant class imbalance in DR datasets, particularly severe underrepresentation of PDR, we hypothesize that synthetic image enrichment will improve sensitivity for minority classes while maintaining overall classification performance.

We designed four experimental setups:

- Experiment 1 (Baseline) uses only the original IDRiD data.
- Experiment 2 (Classical) augments real data with classical augmentation transformations, including rotation, flipping, and brightness adjustment.
- Experiment 3 (GAN) combines real data with GAN-generated images.
- Experiment 4 (Classical + GAN) combines Experiments 2 and 3, using both classical and GAN-generated augmentations together. The synthetic and classical images were sampled equally, such that the total number of augmented images matched that used in each augmentation-only experiment.

After analyzing the class distributions across DR stages in both datasets, we identified the most prevalent classes to exclude from augmentation, focusing augmentation efforts on underrepresented classes. In IDRiD, No DR and Moderate DR were the most prevalent classes, while in FGADR, Severe DR was most prevalent. The majority of classes were not augmented, allowing us to balance the datasets without over-representing already well-represented stages.

The experimental setups were performed with three CNN architectures representing diverse design paradigms: MobileNet V3 Small [57] for lightweight edge deployment, Xception [58] for fine-grained spatial feature extraction, and ConvNeXt Tiny [59] representing modern convolutional architectures.

The evaluation of overall performance and five-class classification (No DR, Mild NPDR, Moderate NPDR, Severe NPDR, PDR) was performed on both datasets. For IDRiD, we used the original test set of 103 images and reserved 10% of the training set for validation. Since FGADR does not provide a publicly available test set, we split the dataset into training (80%), validation (10%), and test (10%) sets. Critically, both test sets consist entirely of real, unseen images with no synthetic augmentation, ensuring that performance improvements reflect genuine generalization rather than overfitting to synthetic image characteristics. The original unbalanced class distributions and the augmentation strategies for both datasets are shown in Figure 3.

Performance is assessed using accuracy (ACC), and Cohen's Kappa (κ):

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (10)$$

where P_o is observed agreement and P_e is expected agreement by chance. Cohen's Kappa is particularly valuable for imbalanced datasets, as it provides a reliable performance measurement beyond accuracy alone.

All models are tested on the IDRiD test set using only real, unseen data. Another set of experiments was performed on a larger FGADR dataset, which was not used for image generation.

Each experimental configuration is repeated five times to reduce variance from random initialization and data splits, with reported results representing averages across these runs.

TABLE 2. Hyperparameter search space for DR classification.

Hyperparameter	Type	Search Range
Batch size	Categorical	8, 16, 32
Number of epochs	Categorical	[5, 10, 15]
Learning rate	Log-uniform	[1e-5, 1e-2]
Frozen layers	Integer	[0, 5]
Optimizer type	Categorical	Adam, AdamW, SGD, Adagrad
Loss function	Categorical	CrossEntropy, FocalLoss, WeightedCrossEntropy
Focal Loss gamma (γ)	Continuous	[0.5, 5.0] (if applicable)

Hyperparameter tuning plays a critical role in optimizing model performance. We employ Optuna [60], an automated optimization framework, to systematically explore a predefined search space for each neural network architecture. Each architecture undergoes independent tuning, ensuring that the optimal configuration is found for its specific characteristics. The tuning process involves 100 trials, each selecting a unique combination of hyperparameters. Performance is evaluated using Cohen's Kappa on the validation set, and the model checkpoint achieving the highest validation Kappa across all training epochs is selected for final evaluation on the test set.

The optimized hyperparameters included the batch size and the number of epochs. The learning rate was sampled on a log scale to explore a wide range of convergence speeds. The number of frozen layers was varied to assess the impact of transfer learning depth on feature extraction. Optimizer type was also tuned, selecting from Adam [61], AdamW [62], SGD [63], and Adagrad [64]. The loss function was chosen from standard CrossEntropy [65] and class-imbalance-aware alternatives: FocalLoss [66] and WeightedCrossEntropy [67]. Table 2 provides a detailed summary of the search space for each hyperparameter.

Table 3 summarizes the optimal training hyperparameters identified through Optuna optimization.

All experiments were implemented using the PyTorch [68] framework and executed on a pair of NVIDIA RTX 3090 GPUs, with image batches distributed across both devices to exploit the combined VRAM.

E. EXPERT VALIDATION OF GENERATED IMAGES

Three ophthalmologists with at least five years of clinical experience conducted detailed visual inspection to verify diagnostic relevance and anatomical accuracy, assessing features such as vessel structure, optic disc appearance, and pathological characteristics.

We conducted a masked evaluation study to assess the visual fidelity of our StyleGAN-generated fundus images. The study included 50 fundus images representing various stages of DR: Healthy, Mild NPDR, Moderate NPDR, Severe NPDR, and PDR. These images comprised 25 real images and 25 synthetic images generated by our StyleGAN2 model.

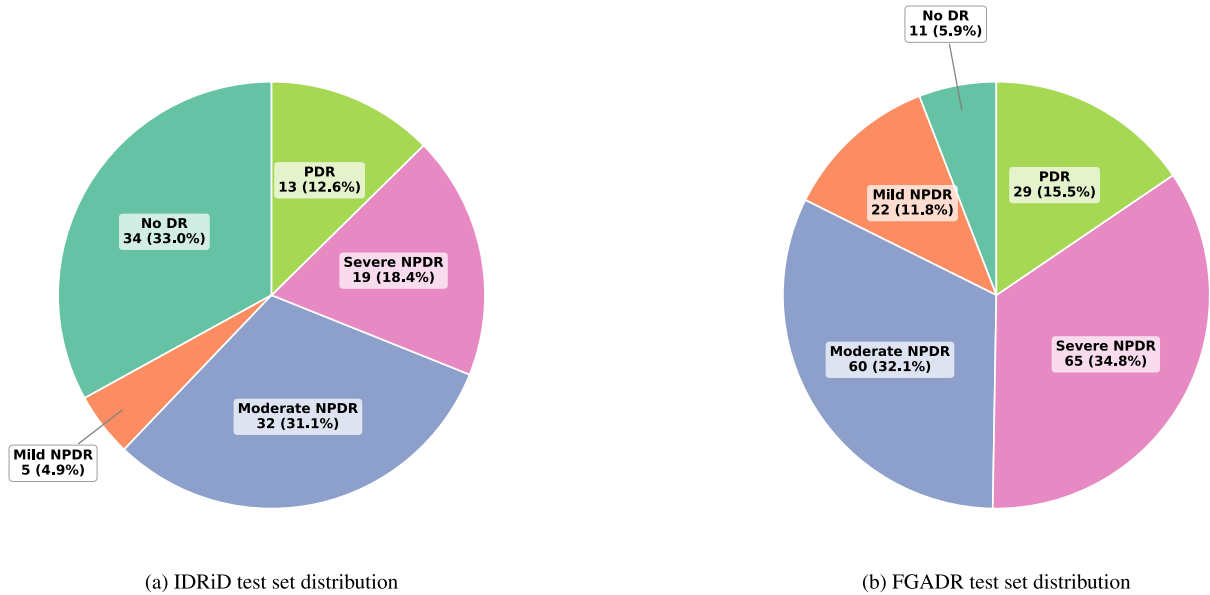


FIGURE 2. Test set class distribution for FGADR (2b) and IDRI (2a) datasets used for 5-class DR classification evaluation.

TABLE 3. Hyperparameter Settings for DR classification experiments.

Architecture	Batch Size	Epochs	Learning Rate	Freeze Layers	Optimizer	Loss Function
ConvNeXt	8	15	2.38e-04	4	Adam	CrossEntropy
MobileNet	8	25	1.00e-03	4	AdamW	CrossEntropy
Xception	16	15	1.14e-03	5	Adagrad	WeightedCrossEntropy

The ophthalmologists were not informed which images were real or synthetic and were instructed to evaluate each image's realism on a 5-point Likert scale:

- 1) Not realistic at all
- 2) Slightly realistic
- 3) Moderately realistic
- 4) Almost real
- 5) Definitely real

This scale was designed to quantify the perceptual gap between synthetic and authentic fundus images. The evaluators were asked to base their assessments on their clinical experience, considering anatomical correctness, vessel structure, lesion appearance, and overall image quality.

In addition to the quantitative score evaluation, ophthalmologists were able to provide qualitative comments explaining their assessment, including the reasons for the assigned score or perceived unrealistic anatomical or pathological features.

IV. RESULTS AND DISCUSSION

This section provides a comprehensive description of the synthetic image generation by our StyleGAN2 model, along with its evaluation. The quality of these images is assessed through both subjective and objective measures. Subjectively,

we conducted a masked study with ophthalmologists to rate the realism of synthetic images. We computed the FID to quantify image fidelity objectively across different DR stages. We also examine the effectiveness of synthetic images as part of the training data in classification experiments, comparing their impact on model performance against and in combination with classical augmentation techniques.

A. SYNTHETIC IMAGE SAMPLES AND TRAINING PERFORMANCE

Examples of synthetic images generated for all dataset classes, together with their corresponding real samples, are presented in Fig. 4. A key advantage of the proposed unconditional StyleGAN-based generation approach is its ability to synthesize complete retinal images without any conditioning input, enabling the generation of an arbitrary number of entirely novel images. Since the generation process is not constrained by external conditions such as masks or partial image inputs, each synthesized image is an entirely new sample that contains no fixed or copied regions from real data. This eliminates the need for lesion masks or additional annotations during generation.

Training five independent StyleGAN2-ADA models, one per DR severity class, required substantial computational

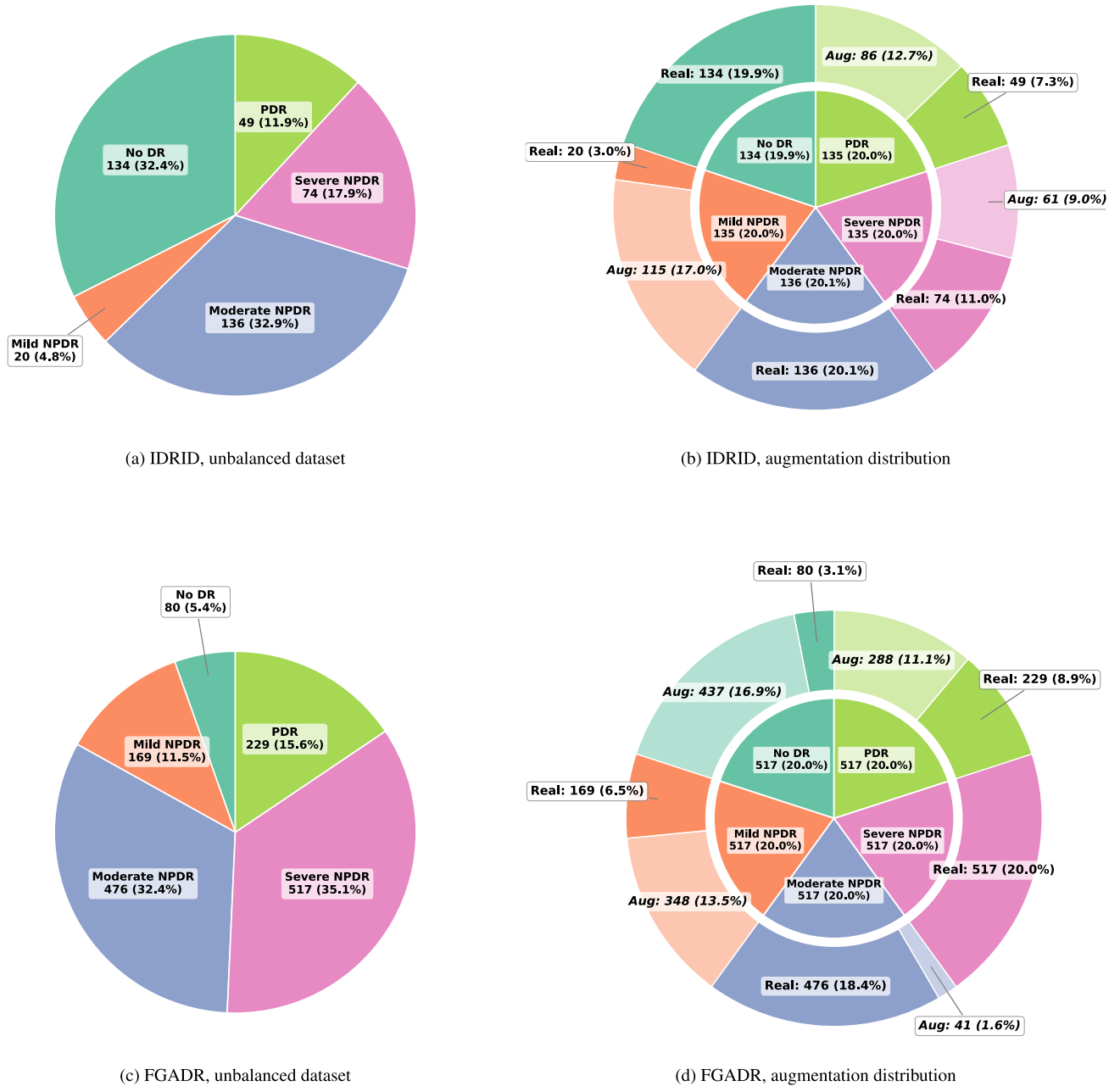


FIGURE 3. Class distributions of the IDRiD 3a and FGADR 3c training sets for 5-class DR classification. The augmentation distributions applied in Experiments 2, 3, and 4 are shown in 3b and 3d.

resources. The overall training process required approximately 1,176 hours, corresponding to 49 days of continuous computation on dual GPUs.

To provide a clear, hardware-agnostic overview of the computational cost, Table 4 summarizes the average training performance across all five models for single-GPU and dual-GPU configurations.

Although the training phase is computationally demanding, image synthesis using the trained generator is highly efficient. For images of resolution 512×512 , the average inference time per image was measured as 0.0594 seconds, enabling near real-time image generation. This fast inference is enabled by the single forward pass of StyleGAN-based generators, a significant advantage over

TABLE 4. Average training performance of StyleGAN2-ADA models under different GPU configurations.

GPU Configuration	Time per king (min)	Training Speed (king/h)	Avg. Training Time per Model (days)
1 GPU	29.2	2.08	12.3
2 GPUs	17.6	3.42	9.8

TABLE 5. Quantitative results per DR stage (50k generated per class). Lower is better for FID and KID (downward arrow); higher is better for IS, SSIM, and PSNR (upward arrow).

DR stage	FID ↓	KID ↓	IS ↑	NN-SSIM ↑	NN-PSNR (dB) ↑
Healthy	49.85	0.031	1.766	0.653	19.65
Mild	46.43	0.021	1.697	0.787	28.14
Moderate	41.58	0.025	1.837	0.715	20.89
Severe	59.57	0.042	1.781	0.674	20.52
Proliferative	70.04	0.041	1.807	0.712	21.30

diffusion-based generative models, which typically require iterative sampling.

B. QUANTITATIVE EVALUATION

We computed the FID between the real and synthetic image distributions across different DR stages to evaluate the objective image quality. FID is a widely used metric for assessing the realism and diversity of generated images, with lower values indicating greater similarity to the real image distribution. As shown in Table 5, the FID scores vary across disease stages, with the best performance observed in the Moderate DR stage (FID = 41.58), and higher values for more complex stages such as Severe (FID = 59.57) and Proliferative DR (FID = 70.04). This suggests that while the model effectively captures key characteristics of healthy images and mild and moderate stages, generating high-fidelity images for advanced DR stages remains more challenging due to increased lesion complexity and structural variability. KID results confirm the FID findings, with the lowest values observed for the Mild (KID = 0.021) and Moderate (KID = 0.025) stages. In contrast, the Severe (KID = 0.042) and Proliferative (KID = 0.041) stages showed higher values. This consistent pattern across both distribution-based metrics confirms that early-stage DR images are generated with higher fidelity than advanced stages.

IS values remained relatively stable across all stages, ranging from 1.697 to 1.837, indicating that the model maintains consistent output diversity regardless of disease severity.

For pixel-level evaluation, we computed NN-SSIM and NN-PSNR between synthetic images and their nearest real counterparts. The Mild stage achieved the highest structural similarity (NN-SSIM = 0.787) and pixel-level fidelity (NN-PSNR = 28.14 dB), substantially outperforming other stages, which ranged from 0.653 to 0.715 and from 19.65 to 21.30 dB, respectively. This strong performance on Mild stage images across all metrics likely reflects the more uniform appearance and less variable lesion patterns characteristic of early DR progression.

C. TASK-BASED EVALUATION OF GENERATED IMAGES

The classification results provide a comprehensive view of the effectiveness of different data augmentation strategies in addressing dataset imbalance, as described in the methodology. The analysis includes class-wise recall, overall accuracy, and Cohen's Kappa, reported for each of the three architectures.

Overall results on the IDRiD and FGADR datasets are presented in Table 6, with per-class TPR shown in Figure 5.

Overall accuracy improved with combined augmentations with the ConvNeXt Tiny model in both datasets. On the other hand, the overall accuracy in both datasets with MobileNet V3 Small was highest when only synthetic augmentations were used. With the Xception model, the gain was slightly more substantial with IDRiD than with the FGADR dataset when synthetic images were used.

In the ConvNeXt Tiny model, combining augmentations yielded the highest Kappa values. Notably, MobileNet V3 Small and Xception achieved the highest Kappa with standard augmentations in the IDRiD dataset, suggesting that the optimal augmentation strategy may depend on both architecture and dataset characteristics. In general, Kappa values were higher in the FGADR dataset than in the IDRiD dataset, most likely because the FGADR dataset is larger.

In the five-class classification task, the results differed significantly between disease stages, models, and datasets. The percentage of augmented images per class may also be a contributing factor. Overall, class-wise improvements were mainly observed in both ConvNeXt Tiny and MobileNet V3 Small when augmentations included synthetic images. These results suggest that synthetic data effectively addressed MobileNet's limitations by enriching the training data's diversity.

The most prominent improvement was the combined augmentation with ConvNeXt Tiny in PDR across both datasets, whereas MobileNet V3 Small and Xception showed the weakest gains. This success in the most severe and heterogeneous PDR class highlights the advantages of the ConvNeXt model architecture. The most significant differences between the IDRiD and the FGADR datasets appeared in the healthy image class. In the IDRiD dataset, this class was among the most prevalent and did not require augmentation to balance the dataset. In contrast, in FGADR this class was the least represented and required substantial augmentation. This result was unexpected because there are no pathological changes in this class, which should make it more easily distinguishable from other classes.

Classification of mild DR failed entirely on the IDRiD dataset due to the critically low number of training images (only 20). None of the augmentation strategies were helpful in this case, including synthetic ones. The challenge with this class is that its few pathological changes are mostly subtle and similar in appearance to healthy images. In the moderate and severe DR classes, there was a substantial effect of synthetic augmentations alone, or in combination with

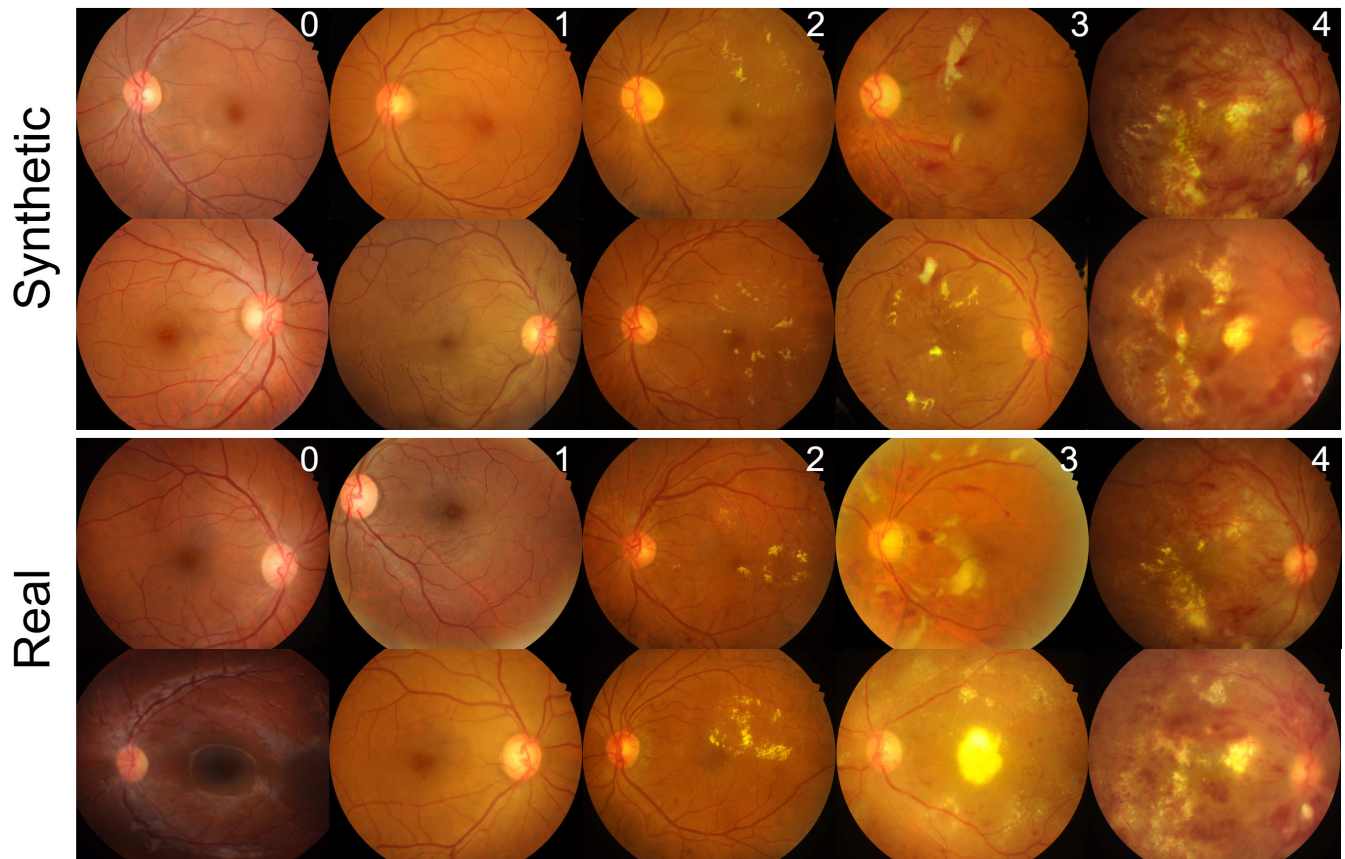


FIGURE 4. Comparison of synthetic fundus images generated by StyleGAN2-ADA (top row) and real fundus images from the IDRiD dataset (bottom row), representing DR stages: 0 - Healthy, 1 - Mild NPDR, 2 - Moderate NPDR, 3 - Severe NPDR, and 4 - PDR.

TABLE 6. Overall results of five-class diabetic retinopathy classification with data augmentation on the IDRiD and FGADR datasets. Acc denotes classification accuracy (%), and κ denotes Cohen's kappa coefficient. Values are reported as mean $\pm$ standard deviation over multiple runs; best results are shown in bold.

Dataset	Exp.	ConvNeXt T		MobileNet V3 S		Xception	
		Acc (%)	κ	Acc (%)	κ	Acc (%)	κ
IDRiD	1	63.69 $\pm$ 3.28	0.496 $\pm$ 0.044	41.36 $\pm$ 5.61	0.213 $\pm$ 0.050	59.03 $\pm$ 3.44	0.423 $\pm$ 0.048
	2	59.81 $\pm$ 5.91	0.454 $\pm$ 0.065	59.61 $\pm$ 2.79	0.444 $\pm$ 0.035	58.83 $\pm$ 4.54	0.428 $\pm$ 0.062
	3	63.30 $\pm$ 3.44	0.485 $\pm$ 0.048	60.00 $\pm$ 1.88	0.439 $\pm$ 0.026	55.73 $\pm$ 2.92	0.378 $\pm$ 0.039
	4	64.85 $\pm$ 4.77	0.515 $\pm$ 0.058	58.45 $\pm$ 2.07	0.423 $\pm$ 0.027	59.09 $\pm$ 2.79	0.422 $\pm$ 0.033
FGADR	1	71.66 $\pm$ 2.27	0.607 $\pm$ 0.032	67.49 $\pm$ 1.95	0.561 $\pm$ 0.025	69.84 $\pm$ 3.31	0.584 $\pm$ 0.046
	2	72.73 $\pm$ 1.17	0.622 $\pm$ 0.020	68.02 $\pm$ 4.09	0.571 $\pm$ 0.048	71.44 $\pm$ 2.21	0.603 $\pm$ 0.034
	3	68.56 $\pm$ 2.56	0.566 $\pm$ 0.028	69.41 $\pm$ 1.33	0.578 $\pm$ 0.020	71.02 $\pm$ 1.74	0.592 $\pm$ 0.026
	4	73.48 $\pm$ 1.74	0.633 $\pm$ 0.026	68.56 $\pm$ 2.20	0.570 $\pm$ 0.024	70.16 $\pm$ 1.70	0.584 $\pm$ 0.026

classical augmentations, for ConvNeXt Tiny and MobileNet V3 Small. This demonstrates that targeted synthetic data generation can enhance model sensitivity to more advanced stages of disease. In contrast, Xception appeared less influenced by the augmentations, except for the moderate DR class, and even experienced a drop when moving from classical to synthetic augmentation.

Overall, GAN-generated data remained highly beneficial as a complementary strategy for diverse and robust training scenarios. GAN-based synthetic augmentation consistently improved performance, especially for models with inherent

limitations, underscoring its value in diverse clinical imaging contexts. The experimental results confirm that data augmentation significantly enhanced model performance in DR classification, particularly when addressing class imbalance. Classical augmentation consistently improved generalization when applied across all classes. GAN-generated data proved particularly valuable in targeted augmentation setups, boosting recall for underrepresented classes and offering meaningful performance gains for lightweight architectures. However, when there are few images per class, even combined augmentations have a limited effect.

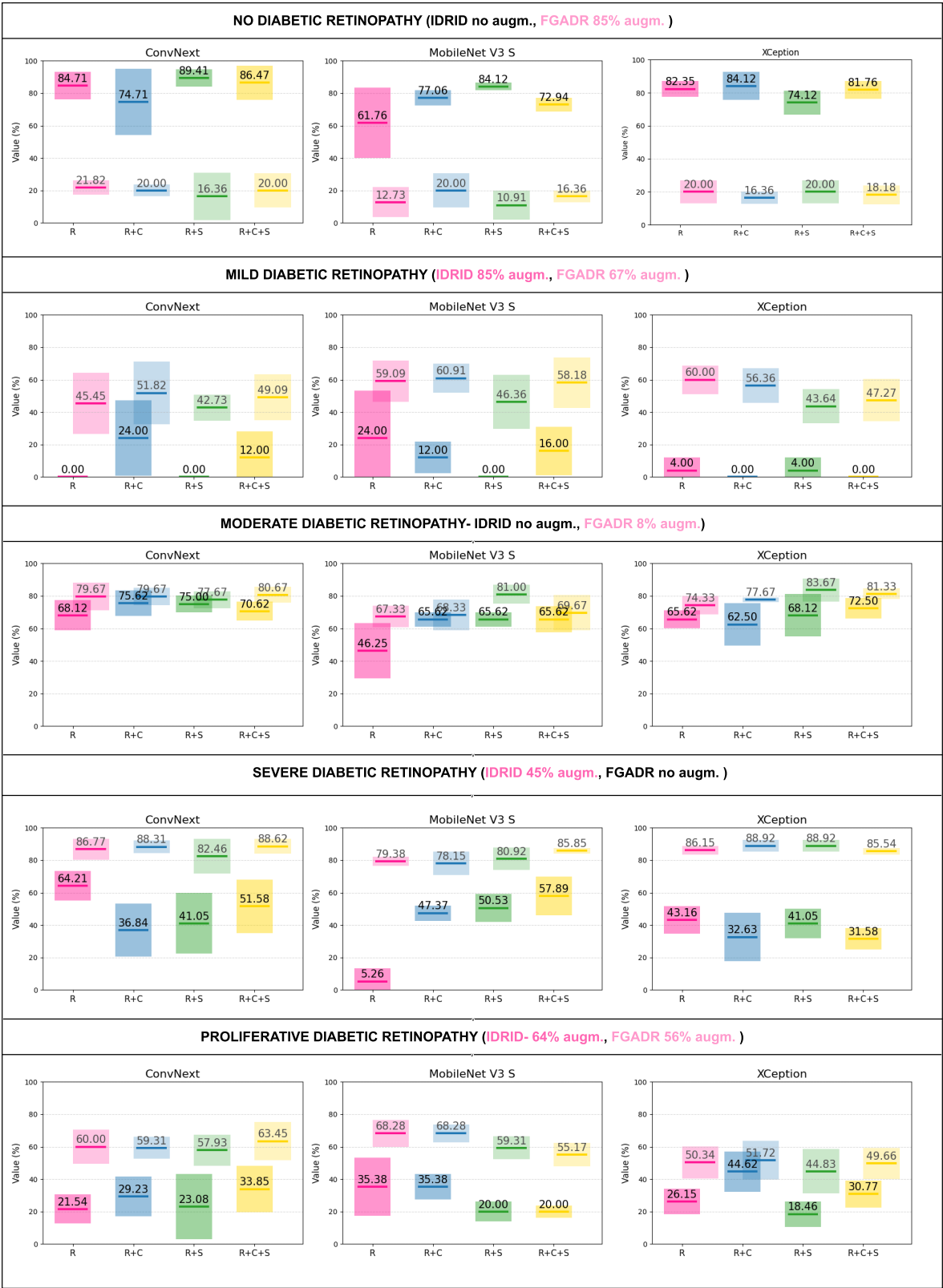


FIGURE 5. True Positive Rate (TPR) per DR class across four experimental conditions (Baseline, Classical augmentation, Synthetic augmentation, and Combined augmentation) for ConvNext Tiny, MobileNet V3 Small, and Xception architectures on IDRID and FGADR datasets.

TABLE 7. Summary of rating performance by evaluators.

Metric	OP1	OP2	OP3
ACC	98.0%	62.0%	88.0%
Sensitivity	96.0%	100.0%	100.0%
Specificity	100.0%	24.0%	76.0%
Positive Predictive Value	100.0%	56.8%	80.6%
Negative Predictive Value	96.2%	100.0%	100.0%
AUC	0.98	0.62	0.88
Mean rating (R)	4.60 $\pm$ 0.71	4.96 $\pm$ 0.20	4.88 $\pm$ 0.33
Mean rating (S)	1.68 $\pm$ 0.69	4.00 $\pm$ 0.71	3.08 $\pm$ 0.70

We hypothesize that the percentage of synthetic images in augmentations per class is crucial. A high percentage, as in No DR in the FGADR dataset (85%) and Mild DR in the IDRiD dataset (85%), resulted in lower sensitivity. The optimal range of synthetic-image percentages in augmentations remains to be investigated, and these values will likely differ depending on the image-generation strategy. Nevertheless, synthetic augmentation must be applied thoughtfully, as excessive or poorly matched generated samples may reduce specificity or cause confusion between adjacent classes. Among the architectures evaluated, ConvNeXt Tiny consistently achieved the best balance of performance.

These findings support the practical value of both classical and synthetic augmentation methods, highlighting the potential of generative models in addressing class imbalance in medical imaging, particularly for real-world screening and diagnostic systems operating under varying data quality and distribution constraints.

D. EXPERT VALIDATION

The evaluation results revealed substantial differences in discrimination capability among the three ophthalmologists, as shown in Table 7. Ophthalmologist 1 (OP1) demonstrated exceptional discriminative ability with the highest accuracy (98.0%) and perfect specificity (100.0%). This evaluator maintained the greatest mean rating difference between real (4.60) and synthetic (1.68) images (2.92 points), suggesting a keen ability to detect subtle artifacts or unrealistic features in generated images. Ophthalmologist 2 (OP2) exhibited perfect sensitivity (100.0%) but very low specificity (24.0%), resulting in the lowest overall accuracy (62.0%). The slight difference between mean ratings for real (4.96) and synthetic images (4.00) indicates a tendency to rate most images as highly realistic regardless of their origin. Ophthalmologist 3 (OP3) achieved a balanced performance with high accuracy (88.0%), perfect sensitivity (100.0%), and reasonable specificity (76.0%). The moderate difference in mean ratings (1.80 points) suggests a more calibrated assessment approach than the other evaluators. The AUC values (0.98, 0.62, and 0.88 for Ophthalmologists 1, 2, and 3, respectively) further quantify the discriminative performance, with OP1 showing near-perfect discrimination and OP2 demonstrating performance only slightly better than random chance. Figure 6 shows the distribution of realism scores assigned by the three ophthalmologists to real and synthetic fundus images. OP1 clearly distinguished between

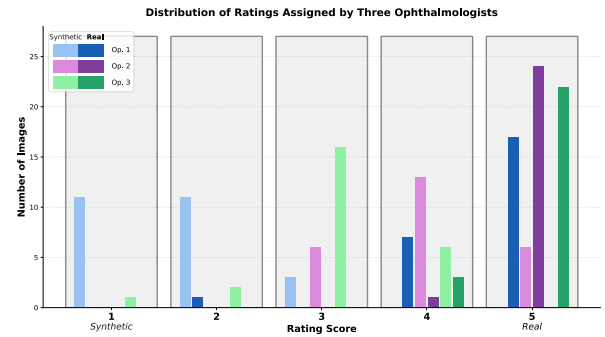
**FIGURE 6.** Distribution of ratings (from 1 - not realistic to 5 - realistic fundus image) assigned by three ophthalmologists (OP1, OP2, and OP3) for both real and synthetic fundus images.

image types, assigning higher scores to real images and lower scores to synthetic images. OP2 rated the majority of images as highly realistic, regardless of authenticity, indicating a higher false-positive tendency. OP3 displayed a moderate pattern, with real images receiving high ratings and synthetic images receiving mid-range scores, suggesting partial differentiation. We calculated inter-rater reliability using Cohen's Kappa coefficient to assess consistency in the ophthalmologists' ratings. The Cohen's Kappa values were computed as follows:

- κ (OP1 vs. OP2) = 0.224 (Fair agreement)
- κ (OP1 vs. OP3) = 0.723 (Substantial agreement)
- κ (OP2 vs. OP3) = 0.364 (Fair agreement)

The substantial agreement between OP1 and OP3 ($\kappa = 0.723$) indicates consistent evaluation standards among experts with strong discriminative ability. The lower agreement involving OP2 reflects divergent assessment strategies rather than poor image quality, as evidenced by OP2's high realism ratings for synthetic images. These kappa values support the clinical plausibility of the generated fundus images. According to the graph in Fig. 6, ophthalmologists can almost exactly identify real images (high number of rating 5), but struggle in the exact identification of synthetic ones (low number of rating 1). We can interpret this as a partial success of our method, which can be further developed based on these evaluation findings and comments.

We performed a detailed analysis of misclassified images to gain deeper insights into the strengths and limitations of our StyleGAN-generated fundus images. Several patterns emerged from this analysis. Generated images in subfigures 7a, 7b, 7c, 7d were consistently misclassified as real by at least two ophthalmologists, representing particular successes in our image synthesis approach. These images exhibit well-formed vascular structures, a realistic optic disc representation, and natural-appearing pathology. OP2 exhibited a strong tendency toward false positives, misclassifying 19 out of 25 generated images as real (76%). This suggests that this evaluator applied different criteria for realism or that our StyleGAN model successfully captured the features this particular expert associated with authentic fundus images. Fig. 7e, a real fundus photograph, was misclassified

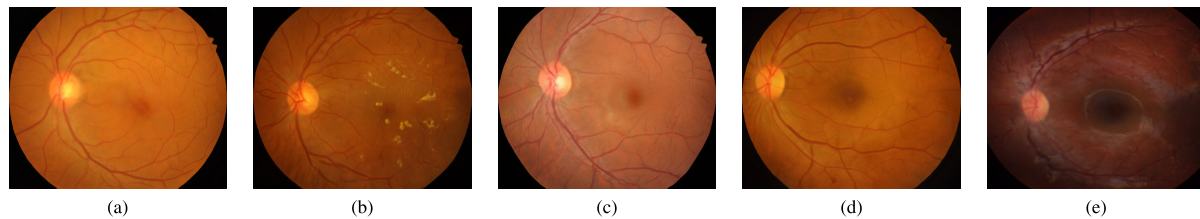


FIGURE 7. Representative misclassified fundus images from expert evaluation. Generated images 7a, 7b, 7c, 7d that were frequently identified as real, and a real image 7e misclassified as synthetic, illustrating the challenges in distinguishing between authentic and StyleGAN-generated fundus photographs.

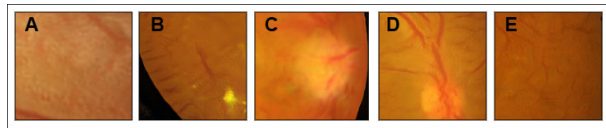


FIGURE 8. Examples of unrealistic features commented by ophthalmologists: unrealistic “patchy” background (A), “strange” parallel lines at the image periphery acting like blood vessels (B), unrealistic optic disc (C), discontinuity of large vessels (D), discontinuity of small vessels (E).

as synthetic by OP1, the otherwise most discriminating evaluator. Detailed examination revealed unusual lighting conditions and vascular patterns that deviated from typical fundus images, highlighting how atypical but genuine clinical images can sometimes appear synthetic to expert evaluators. In addition, ophthalmologists provided qualitative comments on some images, including descriptions of unrealistic features. Examples of such findings are visualized in Fig. 8. These observations provide useful guidance for refining the generation process and improving the clinical fidelity of future synthetic images.

V. CONCLUSION AND FUTURE WORK

In this work, we proposed a GAN-based method for generating synthetic fundus images across different stages of diabetic retinopathy, using separate unconditional StyleGAN2-ADA models for each class. This approach addresses class imbalance without relying on label conditioning or segmentation masks, while capturing key pathological features with high fidelity.

Our experiments demonstrated that synthetic augmentation improved overall classification performance, particularly with the ConvNeXt Tiny architecture. Incorporating synthetic images alongside classical augmentations enabled the ConvNeXt Tiny model to achieve its highest performance on the FGADR dataset, with an accuracy of $73.48\% \pm 1.74\%$ (+1.82% over the baseline) and a Cohen’s Kappa of 0.633 ± 0.026 (+0.026 over the baseline). These improvements were validated by standard metrics and confirmed by ophthalmologists, who found several synthetic images visually indistinguishable from real ones. Notably, ophthalmologists had difficulty reliably identifying synthetic images.

Compared to traditional augmentation methods, our GAN-based approach offers effective, flexible augmentation, making it a practical solution for improving real-world DR screening systems. Although our results are promising, some limitations remain. Advanced DR stages (Severe NPDR,

PDR) exhibit higher FID scores (59.57 and 70.04, respectively), indicating a need to improve pathology diversity. When comparing class-wise results across the IDRiD and FGADR datasets, the optimal proportion of synthetic images in the augmented training data remains unclear, as a high proportion (85%) was associated with lower true-positive rates in some classes. This requires further investigation in future experiments. The results also vary depending on the deep learning architecture used. Finally, although training StyleGAN2-ADA requires substantial computational resources, it remains significantly less computationally demanding than diffusion-based models.

Future work could explore newer generative image models, such as diffusion-based models, to improve quality, develop multi-stage augmentation strategies, or integrate with segmentation-based approaches. This method could be extended to other medical imaging tasks where class imbalance limits model performance, providing a generalizable framework for synthetic data augmentation in clinical applications.

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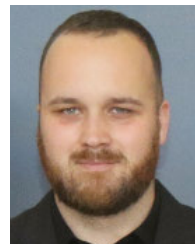
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VERONIKA KURILOVA received the M.D. degree from the Faculty of Medicine, Comenius University Bratislava, the specialization degree in ophthalmology from the Faculty of Medicine, Slovak Medical University in Bratislava, and the Ph.D. degree in applied informatics from Slovak University of Technology, where she focused on applying deep learning methods to ophthalmologic images. She is currently a Doctor and a Specialist Lecturer with the Department of Ophthalmology, Faculty of Medicine, Slovak Medical University in Bratislava. Her research interests include digital signal and image processing, computer vision, artificial intelligence, neural networks, and machine learning. She is a fellow of European Board of Ophthalmology.



learning, and neural networks.

ONDREJ STRAKA received the M.Sc. degree in computer science from the Faculty of Electrical Engineering and Information Technology, in 2022. He is currently pursuing the Ph.D. degree with the Faculty of Electrical Engineering and Information Technology, Slovak University of Technology in Bratislava. His research focuses on computer vision in medicine, especially the segmentation of small anatomical structures. His broader research interests include medical image processing, deep



Bratislava. He has been involved in several AI-based medical imaging projects aimed at enhancing diagnostic accuracy through innovative machine learning techniques. His research interests include the application of deep learning and generative AI in medical diagnostics, with an emphasis on synthetic data generation using GANs to improve the detection of retinal diseases.

MICHAL KOVAC received the bachelor's and master's degrees in robotics and cybernetics from the Faculty of Electrical Engineering and Information Technology, in 2021 and 2023, respectively. He is currently pursuing the Ph.D. degree with the Faculty of Electrical Engineering and Information Technology, Slovak University of Technology in Bratislava. He is a Doctoral Researcher with the Faculty of Electrical Engineering and Information Technology, Slovak University of Technology in

Bratislava. He has been involved in several AI-based medical imaging projects aimed at enhancing diagnostic accuracy through innovative machine learning techniques. His research interests include the application of deep learning and generative AI in medical diagnostics, with an emphasis on synthetic data generation using GANs to improve the detection of retinal diseases.



and SHAP. He has professional experience in medical imaging software development, including radiology tools for CT, MRI, and ultrasound, as well as cloud-based platforms for medical data preprocessing. His work combines applied research in medical image analysis with modern software technologies, including containerized systems and full-stack development. His research interests include medical image AI, XAI, health informatics, and the development of scalable software systems for clinical use.

PETER MACSIK received the M.Sc. degree in software engineering and the Ph.D. degree in cybernetics from the Faculty of Electrical Engineering and Information Technology, Slovak University of Technology (STU) in Bratislava, Slovakia. During his doctoral studies, he focused on developing AI-based diagnostic tools for ophthalmology, with particular emphasis on ensemble learning and explainable artificial intelligence (XAI) methods, including Grad-CAM, LIME,



with the Departments of Ophthalmology, SZU and UNB, Bratislava. She is also an Assistant Lecturer with the Faculty of Medicine, SZU.

ZLATICA FELLNER received the M.D. degree from the Faculty of Medicine, Comenius University Bratislava, in 1999, the dual master's degree in ophthalmology, in 2004 and 2007, the Ph.D. degree in ophthalmology from Charles University, Prague, in 2008, and the M.B.A. degree from Slovak Management Academy, in 2012. She has extensive clinical experience in anterior segment and retinal diseases, with a focus on cataract surgery and anti-VEGF therapy. She is currently



signal and image processing, computer vision, health informatics, and explainable artificial intelligence.

JARMILA PAVLOVICOVA was born in Slovakia. She received the M.Sc. and Ph.D. degrees in telecommunication engineering, in 2002 and 2006, respectively, and the Ph.D. degree in cybernetics from the Faculty of Electrical Engineering and Information Technology, Slovak University of Technology (STU) in Bratislava, Slovakia, in 2015. She is currently the Deputy Director of the Institute of Robotics and Cybernetics, STU in Bratislava. Her research interests include digital

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